

BEN HMIDA AYA

Software Developer |Web, Mobile, IoT

51 325 648 aya.benhmida2001@gmail.com Ben Hmida Aya

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PROFILE

IoT & Web Developer with a Master's degree in IoT Systems Development. Specialized in embedded development using ESP32/Arduino and REST API development. Passionate about connected technologies and motivated to design innovative solutions while growing in a collaborative environment.

EXPERIENCE

Master's Final Project Internship – IoT, Web & Mobile Developer | Feb 2026 – May 2026 | Business Incubator – Mahdia

- Developed a fire detection system using ESP32 and sensors
- Built REST APIs using ASP.NET Core.
- Developed a web interface and mobile application using React and React Native with Expo Go.
- Implemented real-time alerts using SignalR and IoT communication via MQTT/HTTP.
- Used PostgreSQL for data storage and integrated Llama 3.1 (via Groq Cloud) for an AI chatbot.

Summer Internship – Web Development | July 2025 – August 2025 | TSE Consultant INT

- Developed an intern management platform. .
- Created REST APIs using .NET 8 and a front-end application with React.
- Performed API automated testing using Swagger and Selenium.

Final Project Internship – Web Development | Feb 2024 – May 2024 | V5 Agency

- Developed a social platform for startups, created REST APIs using Laravel, built the front-end interface with React, and tested APIs using Postman while managing source code with GitHub.

PROJECTS

SmartWash – Intelligent IoT System for Connected Laundry Services |Oct 2025 – Dec 2025

- Designed and developed a connected solution for real-time washing machine monitoring and reservation.
- Programmed an ESP32 board using PlatformIO to collect machine status data..
- Developed a secure REST API with ASP.NET Core 8 including JWT authentication, payment, and reservation management.
- Built an administration dashboard with React.js and a mobile client application with React Native and Expo.

MES Resource Management – Manufacturing Operations Management Platform | Full-Stack Project

- Developed a collaborative web platform for managing and monitoring manufacturing operations, including machines, operators, stocks, alerts, and breakdowns, using React.js/TypeScript, Node.js/Express.js, and MongoDB. Worked in an Agile/Scrum environment using Jira, Git, and GitHub.
- Analysis and implementation of business processes related to production monitoring, inventory management, machine maintenance and incident management.

EDUCATION

Professional Master's Degree in IoT Systems Development - ISET Mahdia | 2024 – 2026

Bachelor's Degree in Information Technology – DSI – ISET Kélibia | 2021 – 2024
Baccalaureate in Computer Science | JUNE 2021

SKILLS

- **Languages**: C#, Java, PHP, JavaScript, Python
- **Frameworks & Library** : ASP.NET Core (.NET 8), Laravel, Node.js, Express.js, React,React Native, Angular.
- **Databases :** SQL Server, MySQL, MongoDB,PostgreSQL
- **Tools & Testing** : Git, GitHub, Postman, Swagger, Selenium, Trello, SCRUM ,Azure.
- **Concepts** : REST APIs , OOP, Algorithmics, MVC.
- **Business Analysis:** Business Process Analysis, Workflow Modeling, UML, Functional Testing.

LANGUAGES

- **Arabe** : Native
- **Français** : Professional
- **Anglais** : Fluent

SOFT SKILLS

- **Teamwork**
- **Autonomy**
- **Communication skills**
- **Priority Management**